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Problem 1: Texture Analysis

I. Abstract and Motivation

The objective of this section is to explore computer vision techniques for texture classification. Using a dataset consisting of four distinct texture categories (Blanket, Brick, Grass, and Stones), I investigated the efficacy of Laws Texture Energy Measures for feature extraction. Furthermore, I evaluated the impact of feature dimensionality reduction via Principal Component Analysis (PCA) on both unsupervised (K-means) and supervised (Support Vector Machine) machine learning algorithms. The motivation is to understand how spatial frequency information can be linearly transformed and utilized by different classifiers, and to observe the trade-offs between dimensionality reduction and class separability.

II. Approach and Procedures

II..1 Texture Classification - Feature Extraction

To extract texture features, 25 two dimensional 5×5 Laws Filters were generated by computing the tensor products of five 1D kernels: L5 (Level), E5 (Edge), S5 (Spot), W5 (Wave), and R5 (Ripple).

Each of the 48 grayscale images was convolved with these 25 filters using boundary mirroringreflection to handle edge pixels. To compute the energy features, the mean absolute response of each filtered image was calculated, resulting in a 25 dimensional feature vector for each image.

Discriminant power for each feature was analyzed using Fisher's Ratio, which maximizes variance between classes while minimizing variance within class. Following this, Principal Component Analysis (PCA) was applied to the 25-D training features to reduce the dimensionality to a 3-D feature space. Finally, a Nearest-Neighbor classifier was implemented utilizing the Mahalanobis distance to classify the test images based on the 3-D features.

II..2 Advanced Texture Classification - Classifier Exploration

For the advanced classification phase, both unsupervised and supervised learning algorithms were applied to the 25-D and 3-D feature sets to compare performance.

1. **Unsupervised Learning (K-means):** The K-means clustering algorithm was applied to the test images with the hyperparameter K set to 4. The purity of each cluster was evaluated by assigning the cluster to the majority ground truth class within it, and any deviation was counted as an error.

2. **Supervised Learning (SVM):** A Support Vector Machine (SVM) with a linear kernel was trained using the training images. Two separate SVM models were trained(one using the 25-D feature vectors and another using the PCA-reduced 3-D feature vectors). The models were then evaluated on the test set.

III. Experimental Results

III..1 Texture Classification - Feature Extraction

III..1.1 Feature Discriminant Power

Based on the feature extraction pipeline, the discriminant power of the features was quantified using Fisher's Ratio:

- **Strongest Discriminant Dimension:** Dimension 1 (Ratio: 1.30691)
- **Weakest Discriminant Dimension:** Dimension 15 (Ratio: 0.0194643)

Fisher's ratio measures linear separability. Dimension 1 yields the highest ratio, meaning it produces the largest distance between the means of the four texture classes while keeping the variance within each individual class as small as possible. Conversely, Dimension 15 has a ratio near zero, indicating that the energy responses for this specific filter overlap almost entirely across all four classes, rendering it virtually useless for classification.

III..1.2 3-D PCA Feature Space Plot

The 25-D features were reduced to 3-D using PCA. The resulting feature space for the training data is visualized below, demonstrating distinct but partially overlapping clusters.

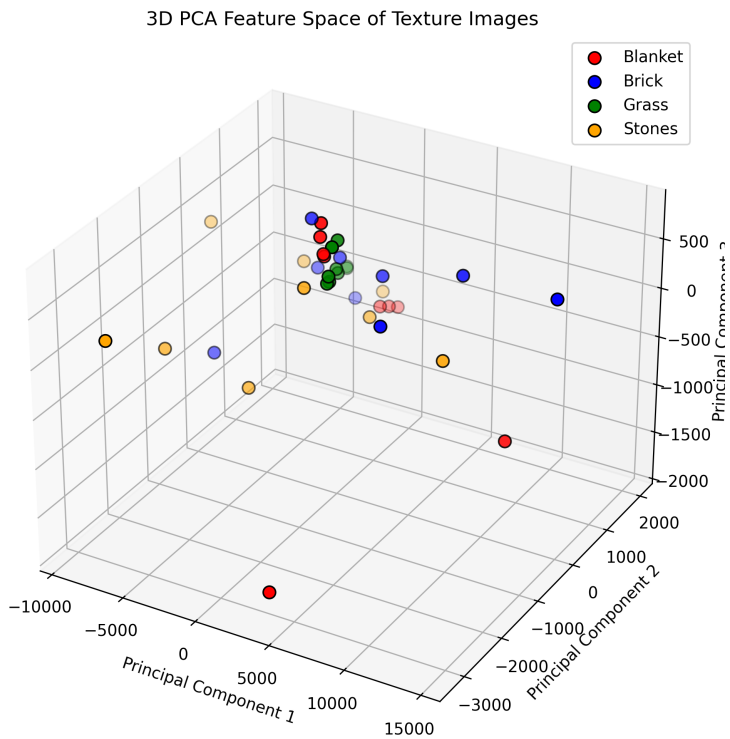


Figure 1: Reduced 3-D feature vectors plotted in the PCA feature space.

III..1.3 Classification Accuracy

The Nearest-Neighbor Mahalanobis classifier was evaluated on the test set. The individual predictions and overall performance metrics are detailed in Table 1 and Table 2.

Test File	Ground Truth	Prediction	Result
1	Grass	Grass	Correct
2	Blanket	Blanket	Correct
3	Blanket	Blanket	Correct
4	Stones	Stones	Correct
5	Stones	Grass	**Incorrect**
6	Grass	Grass	Correct
7	Brick	Blanket	**Incorrect**
8	Stones	Brick	**Incorrect**
9	Brick	Brick	Correct
10	Brick	Brick	Correct
11	Blanket	Stones	**Incorrect**
12	Grass	Grass	Correct

Table 1: Nearest-Neighbor Classification Results (3-D PCA Features)

Metric	Value
Total Test Samples	12
Misclassifications	4
Test Error Rate	33.33%

Table 2: Summary of Classification Performance

III..2 Advanced Texture Classification - Classifier Exploration

The evaluation of advanced classifiers yielded contrasting performances based on the dimensionality of the input data:

Unsupervised Learning (K-Means)

- K-Means Error Rate (25-D Features): **41.67%**
- K-Means Error Rate (3-D Features): **41.67%**

Supervised Learning (SVM)

- SVM Error Rate (25-D Features): **25.00%**
- SVM Error Rate (3-D Features): **66.67%**

IV. Discussion

IV..1 Texture Classification - Feature Extraction

An error rate of 33.33% is reasonable for traditional 5x5 filter banks operating on a heavily compressed 3-D feature space. When comparing the misclassifications with visual observations by eye, the errors align with the limitations of linear filters evaluating highly localized pixel windows:

- **Stones predicted as Grass (Test 5):** In this image, some area contain high frequency, granular components like grass. The Spot(S5) and Edge(E5) filters will yield very similar energy responses for both, causing them to overlap in the feature space.
- **Brick predicted as Blanket (Test 7):** This image has repeating patterns. Moreover, both a brick wall and a woven blanket feature distinct, repeating vertical and horizontal grids. A zoomed out image of brick grout lines can mimic the woven lattice of a blanket to a 5×5 filter matrix.
- **Stones predicted as Brick (Test 8):** Both textures feature repeating, localized block-like structures separated by darker boundaries. For this image, the filters may misinterpret the gaps between stones as the rectilinear joints of a brick wall.

- **Blanket predicted as Stones (Test 11):** For this image, a particularly repeating nubby blanket weave patterns can visually mimicked the varying illumination and depth of small, uniform pebbles.

The 3-D PCA space effectively captures global variance, but visually similar textures inevitably overlap in this constrained 3-dimensional volume, causing the classifier to fail on these specific cases.

IV..2 Advanced Texture Classification - Classifier Exploration

The results from the advanced classifiers highlight a fundamental principle of dimensionality reduction:

- **K-means Observation:** The unsupervised K-means algorithm performed identically (41.67% error) on both 25-D and 3-D datasets. This proves that PCA was effective at retaining the core spatial groupings and overall variance of the data. Because K-means relies on distance based spatial density, stripping 22 dimensions of noise did not negatively impact the fundamental clusters.

- **SVM Observation:** Conversely, comparing it with the unsupervised K-means method, the supervised SVM classifier experienced a massive drop in accuracy when moving from 25-D (25.0% error) to 3-D (66.67% error). PCA is designed to maximize variance, not class separability. By compressing the data down to 3 dimensions, PCA discarded subtle, low-variance features that the SVM required to draw precise discriminative hyperplanes between mathematically similar classes. Therefore, for supervised boundary drawing, retaining the full 25-D feature space was highly advantageous.

Problem 2: Texture Segmentation

I. Abstract and Motivation

The objective of this section is to segment a composite image containing multiple distinct textures into discrete, identifiable regions. Texture segmentation is useful for image parsing, object recognition, and surface inspection. Because textures are characterized by spatial patterns rather than simple pixel intensities, standard thresholding techniques are insufficient. This report outlines an approach using a 2-D Laws' texture energy measures to extract feature vectors for each pixel, followed by unsupervised K-means clustering. Furthermore, using PCA and spatial post processing, it details advanced enhancements implemented to resolve noise and boundary fragmentation inherent in the basic pipeline.

II. Approach and Procedures

II..1 Basic Texture Segmentation

The basic texture segmentation was implemented through the following steps:

1. **Filter Bank Generation:** Five 1-D Laws' kernels of length 5 (L5, E5, S5, W5, R5) were used. The outer products of these 1D vectors were computed to generate twenty-five 5×5 2D filter masks. Mirroring was used for boundary extension.

2. **Convolution & Energy Computation:** The original 512×512 grayscale raw image was convolved with each of the 25 filters using mirror boundary conditions. To compute the texture energy, a 15×15 sliding window was applied to the convolved images, calculating the mean absolute response for each center pixel.

3. **Feature Normalization:** The 25-dimensional feature vectors were normalized to account for global illumination and contrast variations. The energy from the $L5^T L5$ filter represents average local brightness and is generally unhelpful for structural texture classification. Therefore, the remaining 24 features were divided by the $L5^T L5$ energy, and the $L5^T L5$ feature was subsequently discarded, leaving a 24-D feature vector per pixel.

4. **K-Means Clustering:** An unsupervised K-means clustering algorithm ($K = 6$) was applied to the 24-D feature space. Each pixel was assigned a label from 0 to 5, which was then mapped to evenly spaced grayscale values (0, 51, 102, 153, 204, 255) for visualization.

II..2 Advanced Texture Segmentation

To address the shortcomings of the basic texture segmentation, three advanced techniques were integrated:

1. **Dimensionality Reduction with PCA:** The 24-D feature space contains redundant information. Principal Component Analysis (PCA) was applied to project the data down to a 5-D subspace, capturing the most significant variance while eliminating noise.

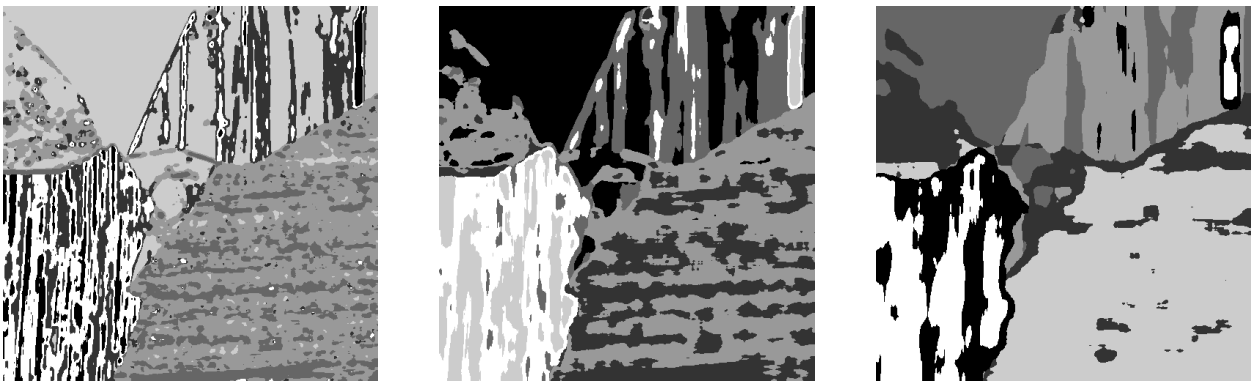
2. **Texture Based Boundary Enhancement:** To resolve blurred boundaries caused by the large energy window overlapping multiple textures, boundary pixels were identified. Their texture energy was computed again using a smaller 5×5 window. These refined, high resolution features were projected through the PCA model, and the pixels were classified again to the nearest K-means centroid.

3. **Post Processing with Majority Filter:** Finally, to correct misclassified standalone pixels(holes) within uniform regions, a spatial Majority(Mode) filter using a 9×9 window was utilized across the clustered label map.

III. Experimental Results

III..1 Basic Texture Segmentation

The output from the basic texture segmentation successfully localized the general layout of the six textures. Among various window size, 15×15 shows the best result. The 15×15 window size is large enough to capture the macro pattern of the textures, resulting in relatively cohesive regions. At the same time, it is small enough that the sharp, geometric boundaries between the different texture wedges in the original mosaic are preserved. However, because of the intrinsic limitation of basic texture segmentation, the image suffered from fragmentation.



(a) 7×7 Window Size (b) 15×15 Window Size (c) 27×27 Window Size

Figure 2: Output of Basic Texture Segmentation using 24-D K-means clustering.

III..2 Advanced Texture Segmentation

The integration of PCA, boundary refinement, and spatial filtering yielded improved results. The regions became solid and cohesive, and the linear boundaries of the original mosaic pattern became sharply defined.

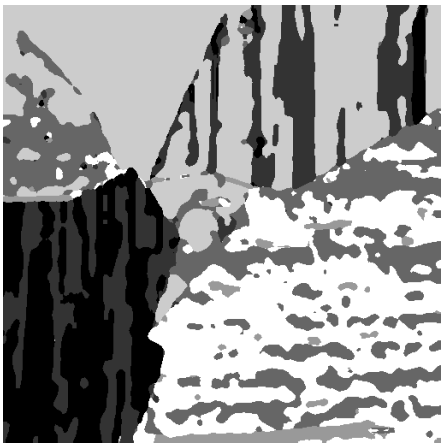


Figure 3: Output of Advanced Texture Segmentation.

IV. Discussion

IV..1 Basic Texture Segmentation

The noise observed in the basic segmentation results from natural internal class variance. Because K-means operates on the 24-D energy vector of an isolated pixel, it possesses no spatial context. A dark knot in a piece of wood or a deep shadow in a heavily grooved texture yields a unique energy signature, causing the algorithm to misclassify it as a distinct texture, resulting in the salt and pepper noise. Furthermore, maintaining all 24 dimensions allows highly specific micro variations to heavily influence the Euclidean distance calculations during clustering.

IV..2 Advanced Texture Segmentation

The advanced implementation proved highly effective by solving three distinct issues.

First, PCA functioned as a feature selector. By reducing the data to 5 dimensions, the algorithm ignored the low variance noise components that were causing confusion during the K-means.

Second, the boundary refinement step solved the windowing effect issues. When a large 15×15 window slides over the edge between two regions, the resulting energy is an ambiguous blend of both textures, causing jagged borders. By identifying those border pixels and evaluating them again with a tight 5×5 window, the algorithm successfully snapped the boundary lines to their true edges based on pure texture properties.

Finally, the 9×9 majority filter introduced necessary spatial awareness. By forcing pixels to fit in the dominant class of their surrounding neighborhood, small remaining misclassified holes were swallowed by the correct larger region, resulting in a clean, highly valid segmentation map.

Problem 3: SIFT and Image Matching

I. Abstract and Motivation

The Scale-Invariant Feature Transform (SIFT) algorithm is widely utilized due to its invariance to image scaling, translation, and rotation, as well as its partial robustness to illumination changes and affine or 3D projections. In this section, first, I evaluated SIFT's performance in direct image matching across different viewing angles and object classes. Second, I constructed a Bag of Words model using K-means clustering to represent images as visual frequency histograms. By analyzing both local feature correspondences and global visual word distributions, we will understand the strengths and limitations of these techniques.

II. Approach and Procedures

II..1 Salient Point Descriptor

The Scale-Invariant Feature Transform (SIFT) properties are summarized as follows:

- **Geometric Robustness:** SIFT is invariant to image translation, scaling, and rotation, and robust to moderate affine distortions and 3D viewpoint changes.
- **Mechanisms of Robustness:**
  - **Translation:** Descriptors are anchored to precisely localized spatial extrema.
  - **Scaling:** Keypoints are extracted as extrema across a continuous scale space.
  - **Rotation:** The descriptor coordinate system is aligned with the keypoint's dominant local gradient orientation.
  - **Affine/3D Viewpoint:** Spatial pooling into histograms allows gradients to shift slightly without altering the descriptor signature.
- **Robustness to Illumination Change:** SIFT uses gradients (canceling additive brightness shifts) and normalizes the descriptor vector to unit length (canceling multiplicative contrast changes). Non-linear changes are handled by thresholding vector values at 0.2 and re-normalizing.
- **DoG vs. LoG Efficiency:** The Difference of Gaussians (DoG) efficiently approximates the computationally expensive Laplacian of Gaussian (LoG) by subtracting adjacent Gaussian blurred images in the scale space pyramid.
- **Output Dimensionality:** The final descriptor is a 128-dimensional vector (came from a 4×4 grid of 8-bin gradient orientation histograms).

II..2 Image Matching

For the basic image matching task, SIFT features (keypoints and descriptors) were extracted from four images: Cat_1, Cat_2, Cat_3, and Dog_1.

1. **Largest Scale Analysis:** The keypoint with the maximum scale value in Cat_1 was isolated. Its nearest neighbor in Cat_3 was found using a brute force matcher with L2 distance.
2. **Pairwise Matching:** Feature matching was performed across four specific pairs using the K-Nearest Neighbors (KNN) algorithm ($k = 2$). To filter out ambiguous matches, Lowe's Ratio Test was applied with a threshold of 0.75.

II..3 Bag of Words

To implement the Bag-of-Words model, all SIFT descriptors from the four images were pooled into a single dataset.

1. **Codebook Generation:** K-means clustering was applied to the pooled descriptors to generate a vocabulary of $k = 8$. The centroid in each cluster served as the codewords.

2. **Histogram Creation:** Each image's descriptors were mapped to their nearest word, producing an 8-bin normalized histogram for each image.

3. **Similarity Measurement:** The Euclidean L2 distance was calculated between the histogram of Cat_3 and the histograms of the other images to evaluate similarity in the BoW space.

III. Experimental Results

III..1 Image Matching

Largest Keypoint Analysis: The SIFT detector extracted 942 keypoints from Cat_1 and 287 keypoints from Cat_3. The keypoint with the maximum scale in Cat_1 was isolated, and its nearest neighbor in Cat_3 was identified.

- **Cat_1 Largest Keypoint:** Scale = 81.4057, Orientation = 343.603°
- **Cat_3 Matched Neighbor:** Scale = 2.58894, Orientation = 74.038°

Pairwise Matching using Lowe's Ratio Test:
Table 3 summarizes the number of robust correspondences found after applying the ratio test threshold.

Image 1	Image 2	Matches (After Ratio Test)
Cat_1	Cat_3	10
Dog_1	Cat_3	6
Cat_3	Cat_2	2
Cat_1	Dog_1	2

Table 3: Number of valid SIFT matches between image pairs.

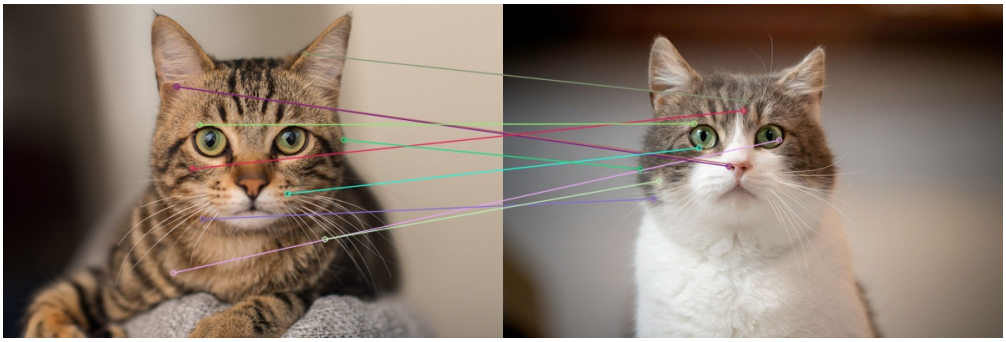


Figure 4: SIFT matches between Cat_1 and Cat_3. 10 matches are found, primarily concentrated around highly structured areas like the eyes and nose.

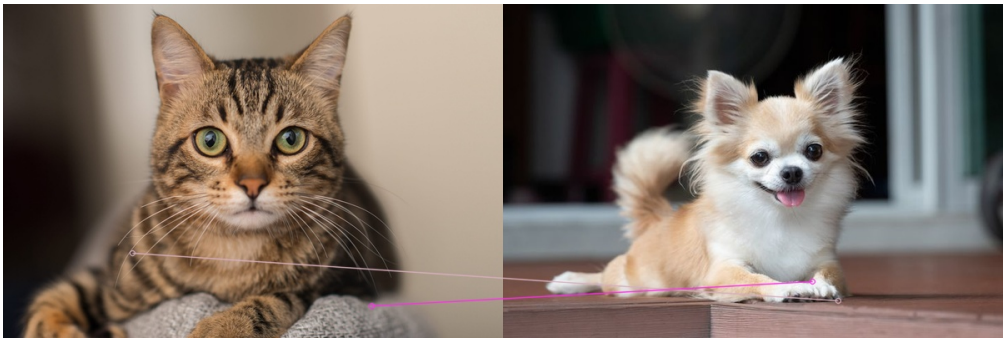


Figure 5: SIFT matches between Cat_1 and Dog_1. Only 2 matches are identified.

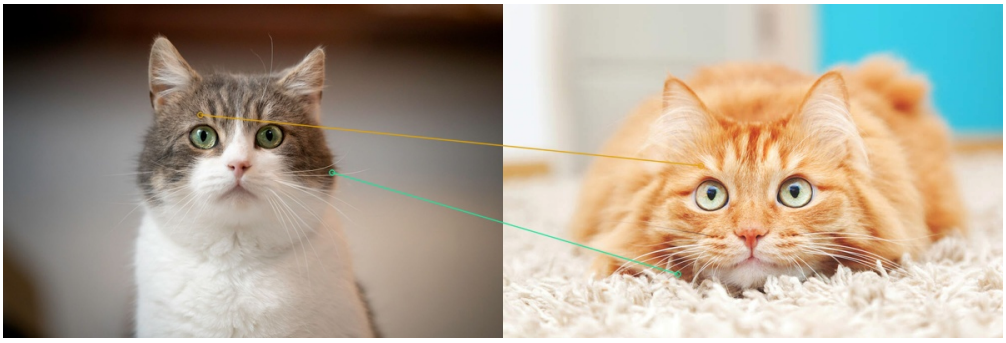


Figure 6: SIFT matches between Cat_3 and Cat_2. Only 2 matches are found even though both subjects are cats.

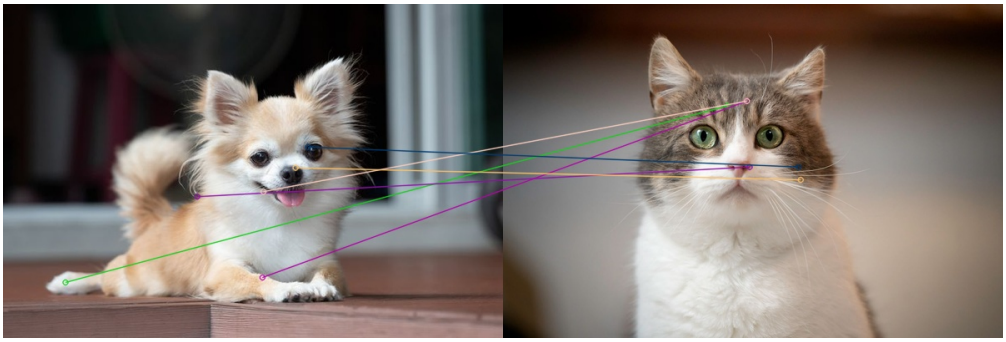


Figure 7: SIFT matches between Dog_1 and Cat_3. 6 matches occur linking unrelated parts.

III..2 Bag of Words

A total of 2,098 SIFT descriptors were extracted across all four images and pooled to build the visual vocabulary. The K-means clustering algorithm ($k = 8$) produced the normalized histograms detailed in Table 4.

Image	Bin 1	Bin 2	Bin 3	Bin 4	Bin 5	Bin 6	Bin 7	Bin 8
Cat_1	0.1221	0.0998	0.1444	0.1125	0.1030	0.1221	0.1741	0.1221
Cat_2	0.1569	0.1405	0.1460	0.1241	0.1332	0.1113	0.1058	0.0821
Cat_3	0.1742	0.1882	0.0941	0.1010	0.0557	0.1115	0.1324	0.1429
Dog_1	0.1215	0.3146	0.0623	0.1059	0.0935	0.1059	0.0779	0.1184

Table 4: Normalized 8-bin Bag of Words histograms for each image.

To evaluate similarity, the Euclidean L2 distances were calculated between Cat_3 and the other images:

- Distance (Cat_3, Cat_2) = 0.1272
- Distance (Cat_3, Cat_1) = 0.1330
- Distance (Cat_3, Dog_1) = 0.1576

IV. Discussion

IV..1 Image Matching

The visual results demonstrate the strength and limitations of the SIFT algorithm.

Orientation and Scale Limitations: SIFT matched a massive keypoint in Cat_1(81.4) to a small keypoint in Cat_3(2.5), with different orientations (343.6° vs 74.0°). This discrepancy highlights that because the images contain entirely different cats, the nearest neighbor match is not physically meaningful. SIFT is simply linking two mathematically similar localized gradients.

Why it Works or Fails in Some Cases: As seen in the provided figures, SIFT struggled significantly across almost all pairs. This is expected because SIFT is designed for instance matching to find the exact same object under different lighting, scale, or slight 3D rotation, not category matching.

- **Matching Different Objects (Cat vs. Dog):** As seen in Figures 5 and 7, matching between fundamentally different objects fails completely. The local textures and geometric structures are too dissimilar. The few lines generated are entirely false positives, as they pass the ratio test by coincidence.
- **Intra-class Variations (Different Cats):** Figures 4 and 6 show two different results. Between Cat_1 and Cat_3, SIFT found about a dozen matches centered around the eyes and nose. Because all cats share similar high-contrast eye structures, SIFT found localized patches of pixels that looked identical. However, between Cat_3 and Cat_2 (Figure 6), it failed. This is because Cat_2 has long orange fur and is in a different pose, whereas Cat_3 has short white and grey fur. The localized gradients are completely different.
- **Viewing Angles:** SIFT handles moderate affine transformations well, but large viewing angle differences alter the visible local textures. If an object rotates too far, the 2D projection of its 3D texture changes beyond what SIFT's local histograms can handle, leading to the sparse matching.

IV..2 Bag of Words

The Bag of Words offers a different approach by discarding spatial relationships in favor of global frequency distributions. The empirical results demonstrate both the strength and the limitations of this method.

Successful Class Separation: Despite using a small codebook size ($k = 8$), the BoW successfully distinguished between intra-class (cat vs. cat) and inter-class (cat vs. dog) relationships. The L2 distances followed the visual similarity: Cat_3 is mathematically closest to Cat_2 (0.1272) and Cat_1 (0.1330). The distance to Dog_1 is the highest (0.1576). This indicates that the images of the cats share a more similar overall distribution of localized gradient patterns than the dog.

Histogram Analysis: A closer look at the raw histograms in Table 4 reveals why the dog was successfully isolated mathematically. In Bin 2, Dog_1 exhibits a spike in frequency(0.3146), whereas the cats maintain much lower frequencies for that same visual word. This suggests that a specific textural feature dominated the K-means cluster for that bin, providing a unique numerical signature that distanced it from the cats.

Vocabulary Size: While the class separation succeeded in this small sample, the limitation of $k = 8$ is evident in the relatively small margins between the distances (the difference between the closest cat and the dog is only 0.03). For better implementation, vocabularies might contain hundreds or thousands of words. With only 8 bins, highly distinct visual features are forcibly clustered together, creating a quantization that reduces the overall discriminative power of the model.